

[!] SUPERSEDED - DO NOT BUY FROM THIS LIST Use docs/MASTER_BOM.md instead. This file is kept only for history. At least two entries here are now WRONG and would waste money: * **4mm pen-style springs** - ruled out permanently in v7.1. Braille rows are 2.6mm apart, so a 4mm spring fouls its neighbour. You need **2mm OD micro springs**. * **2mm bearing balls** - no longer needed at all since v7.1. The braille dot is printed as a dome on the linkage itself; there is nothing to glue a ball onto.

Braillix v5.1 - Complete Electronics & Hardware Shopping List

Assuming: 1 Brain Pod + 1 Cell (prototype). Multiply cell items by N for more cells.

A. ESP32 BRAIN POD

#	Item	Specs	Qty	Est. Price	Source
1	ESP32 DevKit V1-compatible (30-pin)	WiFi+BT, USB-C; use the owned board	1	Already owned	-
2	Female header strip 1x15	2.54mm pitch, through-hole	2	Rs 10 each	Local / Robocraze
3	5.5x2.1mm inline female barrel pigtail	Use the owned cable; not a panel-mount jack	1	Already owned	-
4	5V 3A DC adapter	5.5x2.1mm barrel plug, regulated	1	Rs 200	Local / Amazon
5	6x6x5mm tactile push switch	4-pin DIP, momentary	3	Rs 5 each	Robocraze / LCSC
6	4.7k ohm resistor (1/4W)	Through-hole or 0805 SMD	2	Rs 1 each	Local
7	Piezo buzzer (optional)	5V, 12mm	1	Rs 15	Local

Pod subtotal: ~Rs 420

B. PER CELL - Motor + Mechanical

#	Item	Specs	Qty/cell	Est. Price	Source
8	28BYJ-48 stepper motor	5V DC, 4-phase, 64:1 gearbox	1	Rs 80	Robocraze
9	SS49E Hall effect sensor	Linear, SIP-3, 3-6V	1	Rs 25	Robocraze / LCSC
10	3x1mm NeFeB disc magnet	Neodymium, for cam homing	1	Rs 5	Local / AliExpress
11	2mm stainless steel bearing balls	For braille dot caps	6	Rs 2 each	Local hardware / Amazon
12	Micro compression spring	OD 4-4.5mm, length 5-6mm	6	Rs 3 each	Local hardware

Per-cell mechanical subtotal: ~Rs 140

C. PER CELL - Muscle Board PCB Components

Option A: Get PCBs fabricated (recommended)

#	Item	Qty/cell	Est. Price
13	Muscle board PCB (34x44mm, 2-layer)	1	Rs 35 (JLCPCB, 5pc MOQ)

Option B: Use breadboard/perfboard for prototype

#	Item	Qty/cell	Est. Price
13b	Arduino Pro Mini 5V/16MHz	1	Rs 120
13c	ULN2003 driver board (module)	1	Rs 40

Skip to section D if using Option B

If fabricating PCBs, you need these SMD components per board:

#	Item	Package	Qty/board	Est. Price	LCSC Part
14	ATmega328P-AU	TQFP-32	1	Rs 210	C14877
15	ULN2003A	SOIC-16	1	Rs 25	C7386
16	16MHz crystal	HC49-4H	1	Rs 12	C32346
17	22pF capacitor	0402	2	Rs 1 each	Generic
18	100nF capacitor	0402	2	Rs 1 each	Generic
19	100uF/10V electrolytic	SMD 5x5.3mm	1	Rs 12	C131307
20	10k ohm resistor	0402	2	Rs 1 each	Generic
21	4.7k ohm resistor	0402	2	Rs 1 each	Generic
22	SS14 Schottky diode	SMA	1	Rs 4	C2480
23	1x04 pin header	2.54mm	1	Rs 3	Cut from strip
24	JST XH 5-pin connector	B5B-XH-A	1	Rs 8	C2316
25	1x03 pin header	2.54mm	1	Rs 2	Cut from strip
26	2x03 pin header (ISP)	2.54mm	1	Rs 3	Cut from strip

Per-cell PCB subtotal: ~Rs 320 (custom) or Rs 160 (Pro Mini + ULN2003 module)

D. DAISY-CHAIN / CONNECTORS

#	Item	Specs	Qty	Est. Price	Source
27	4-pin pogo connector (spring+pad pair)	2mm pitch, spring-loaded	N+1 pairs	Rs 40/pair	LCSC / AliExpress
28	8x1mm NeFeB disc magnets (docking)	2 per docking face; 10 owned covers pod + two cells	10	Already owned	-

Count your magnets carefully:

- Each cell: 3 on -X face + 3 on +X face = **6 per cell**
- Pod: 3 on +X dock face = **3 for pod**
- Total for 1 pod + N cells: **3 + 6N magnets**
- For 1 pod + 1 cell prototype: **9 magnets**

E. HARDWARE / FASTENERS

#	Item	Specs	Qty	Est. Price	Source
29	M2.5 x 25mm bolts	Pan head or socket head	4 per cell	Rs 2 each	Local hardware
30	M2 x 6mm screws	Pan head	4/cell (board) + 2 (pod lid)	Rs 1 each	Local hardware
31	M4 bolts + nuts	~10mm, for motor mounting	2 per cell	Rs 2 each	Local hardware
32	M2 grub/set screw	3-4mm long (if using comb later)	1 per cell	Rs 2	Local hardware

F. WIRE / SOLDER SUPPLIES

#	Item	Specs	Qty	Est. Price	Source
33	Hookup wire (stranded)	22-26 AWG, multiple colors	1 roll each: Red, Black, Blue, Yellow, Green, White	Rs 30/roll	Local
34	Heat shrink tubing	Assorted 1-3mm	1 pack	Rs 30	Local
35	Solder wire	60/40 or lead-free, 0.5-0.8mm	1 roll	Rs 80	Local
36	Flux paste	For SMD soldering	1 syringe	Rs 50	Local
37	Superglue (CA)	For bearing balls onto linkage nubs	1 tube	Rs 20	Local
38	Silicone grease	For linkage feet on cam tracks	1 small tube	Rs 40	Local

G. TOOLS (if you don't have them)

#	Item	Why	Est. Price
39	Soldering iron (temp-controlled)	SMD + through-hole	Rs 500+
40	AVR ISP programmer (USBasp)	To flash ATmega328P bootloader	Rs 150
41	Multimeter	Continuity, voltage checks	Rs 200+
42	Wire stripper	22-26 AWG	Rs 100
43	Small file set	Trim motor shaft to 4mm	Rs 80
44	Tweezers (fine tip)	SMD placement + spring insertion	Rs 50
45	Calipers (digital)	Measure shaft, pin pitch, etc.	Rs 300

COST SUMMARY (1 Pod + 1 Cell prototype)

Category	Cost
Brain Pod electronics	Rs 420
Cell mechanical	Rs 140
Cell electronics (Pro Mini route)	Rs 160
Connectors + magnets	Rs 85
Hardware/fasteners	Rs 30
Wire/solder/glue	Rs 250
TOTAL (excluding tools + 3D printing)	~Rs 1,085

For 5 cells + 1 pod: ~Rs 2,500 (well under 15,000 INR budget)

QUICK SHOP LIST (just the names, for the counter)

ESP32 DevKit V1 (30-pin) x1
28BYJ-48 stepper motor x1
Arduino Pro Mini 5V/16MHz x1 (or ATmega328P-AU if doing custom PCB)
ULN2003 driver board x1 (or ULN2003A S0IC-16 if custom PCB)
SS49E Hall sensor x1
5V 3A DC adapter with barrel jack x1
Panel-mount barrel jack 5.5x2.1mm x1
6x6x5mm tactile switch x3
4-pin pogo connector pair x2
3x1mm homing magnet x1; 8x1mm docking magnets x10 already owned
2mm bearing balls x6
Micro compression springs (OD 4mm, 5mm long) x6
Female header strip 1x15 x2
Male header strip 40-pin x2 (cut to size)
JST XH 5-pin connector x1

Resistors: 4.7k x4, 10k x2
22pF capacitor x2, 100nF x2, 100uF/10V x1
SS14 Schottky diode x1
16MHz crystal x1
M2.5x25 bolts x4
M2x8 screws x2 (pod lid); M2x6 x4 only if the deferred muscle board is built
M4x5 thread-forming screws x2; no nuts, no M4x6/x10
Hookup wire (Red, Black, Blue, Yellow, Green, White)
Heat shrink assorted
Solder wire
Superglue
Silicone grease